Jacoti Forsyth

linkedin.com/in/jacotiforsyth ■ jacoti.forsyth@gmail.com ■ (978) 496-6950 ■ Acton, MA

R&D engineer experienced in 3D modeling, product development, process optimization, and operations committed to driving innovation through novel design, processing, and manufacturing techniques.

PROFESSIONAL EXPERIENCE

Lightfab | Engineering Consultant | New Ipswich, NH | 2025 - Present
- Developed and validated designs suitable for novel laser-printing technology
- Commissioned a new facility setting up machines, manufacturing areas, and validation process within 3 months

SpectraWAVE | Research and Development Intern | Bedford, MA | 2023
- Conducted market study to enhance our product viability, afterwards lead a team of engineers to correct design flaws and further enhance product
- Designed, prototyped, and 3D printed fixtures and components to improve manufacturing efficiency and standardize the process for reproducibility
- Designed a display kit that enabled portable and lightweight display of product functionality at conventions and conferences

Creative Materials Incorporated | R&D Engineer | Ayer, MA | 2022
- Formulated specialty material, validating and modifying iteratively to align with customer requirements
- Lead jet dispensing projects using tailored conductive materials and designs to fit their product and application.

ACADEMIC EXPERIENCE

University of Massachusetts Lowell | Graduate Researcher | Lowell, MA |2024 - 2025
- Introduced new materials in 3D printing applications to advance the capabilities of additive manufacturing
- Designed and prototyped application geometry for material and printing validation
- Presented at research convention: Enabling FFF 3D Printing for High-Performance and Functional Materials

Dog Whistle Product Design | Capstone Project | 2023 - 2024
- Led a team of 4 to redesign whistle for dog competitions, increasing distance and efficacy by 10x
- Utilized 3D printing with specialty printers (Formlabs and Markforged) to prototype and refine designs

River hawk key chains | Mold Design Project | 2023
- Led team of 4 in the complete design, simulation, prototyping and production of multifunctional key chains
- Utilized overmold technology with multiple material types to yield highly functional and aesthetic product

SKILLS

Software: SolidWorks, BLENDER, Ansys, MoldX3D, Moldflow, Mastercam, Microsoft Office
Plastics Processing: Injection molding, Dual material filament extrusion, Extrusion, Thermoforming, Compression molding, Rotomolding, Blow Molding
Production: FFF printers, SLA printers, CNC, DSC, TGA, Jet Dispensing

EDUCATION

Master of Science, Plastics Engineering | University of Massachusetts, Lowell | GPA 3.85 | 2025
Bachelor of Science, Plastics Engineering | University of Massachusetts, Lowell | GPA 3.78 | 2024
Relevant coursework: Rapid Prototyping, Computer Aided Engineering, Product Design, Synthetic Fibers, Mechanical Behavior

HOBBIES & INTERESTS

- 3D modeling for game assets and utility prints around the house, Game design development
- Staying active through swimming, diving, volleyball, and pickle ball